Low-Dimensional, Stable, and Moderately Discriminative Subspaces for Engine Sound Attributes

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Abstract

We ask whether engine sounds sharing a fixed attribute live in low-dimensional, stable, and moderately discriminative linear subspaces. Using 5-fold cross-validation on `engine_configuration`, we extract MFCC features ($20 + \Delta + \Delta\Delta$, so $D = 60$), fit class-conditional PCA subspaces with a uniform rank $r = 5$ on TRAIN frames, quantify stability via bootstrapped principal angles, and evaluate discriminativeness with a calibrated nearest-subspace classifier (NSC) using trimmed aggregation ($q = 0.40$, $K \geq 10$).

At $r = 5$, cumulative explained variance is 92 % to 94 % across classes, supporting a low-rank structure. Bootstrapped self-angles are modest (medians ≈13°–15°; inline-6 weaker), indicating subspace stability; between-class largest principal angles range ≈25°–55°, indicating partial overlap. NSC attains 25.0 % (7.1) accuracy versus a 20 % chance baseline; a one-sided permutation test combined across folds via Fisher's method (Eq. 16) yields $\chi^2_{10} = 20.44$, $p = 0.0253$, confirming a statistically significant but practically small lift. Misclassifications track geometry: the closest class pairs in angle space account for most errors.

Implication. Low-rank, reasonably stable subspaces enable compact indexing, robust similarity search, and interpretable diagnostics for engine-audio analytics, while the modest lift cautions against using subspaces alone for fine-grained classification.

Keywords: Audio representation, MFCC, PCA subspaces, principal angles, classification, robustness, bootstrapping.

1 Motivation & Conceptual Framing

Why subspaces for engine sounds? Engine configurations (e.g., inline vs. V-block) determine firing order, cylinder count, and exhaust manifold geometry, which in turn shape periodicity, harmonic spacing, and formant-like spectral envelopes in recorded audio. Despite noise and recording variance, clips from the same configuration should concentrate around a low-dimensional manifold of timbral patterns. Local linear approximations of such manifolds are *class subspaces*.

What do we gain? (i) *Compactness:* Low rank ($r \ll D$) yields memory- and compute-efficient representations for large audio libraries. (ii) *Stability:* If subspaces are reproducible under resampling, they capture configuration-level structure rather than incidental clip idiosyncrasies. (iii) *Interpretability:* Subspace bases (PCA loadings) act like timbral modes; principal angles quantify between-class separations. (iv) *Downstream utility:* Stable, compact subspaces support indexing/retrieval, coarse attribute tagging, and serve as priors for more flexible models (e.g., mixture-of-subspaces, factor models).

This study treats *engine configuration* as a physically grounded attribute and tests three claims: (i) *low-dimensionality* (high EVR at small r), (ii) *stability* (small bootstrap principal angles), and (iii) *moderate discriminativeness* (NSC > chance) — acknowledging that overlapping acoustic manifolds and recording heterogeneity limit separability.

2 Overview

Goal: Test whether engine sounds of a shared attribute lie in low-dimensional, stable, and moderately discriminative subspaces.

Attribute(s): `engine_configuration` (classes: V6, V8, inline-4, inline-6, single-cylinder).

Pipeline (from code): Feature extraction: per-clip MFCC with deltas (D =60) from frames uniformly selected per clip; per-clip CMVN by centering in the subspace pipeline. Subspaces: per-class PCA on TRAIN frames, uniform rank r =5. Stability: bootstrap re-fitting on TRAIN ($B = 1000$ bootstraps, 70 % of clips each) and reporting largest principal angles (degrees). Classification (NSC): frame residuals to class subspaces → *trimmed aggregation* per clip ($q = 0.40$, one-sided upper-tail trim unless $K < 10$; fallback to median) → per-class z-score calibration estimated on TRAIN → argmin on calibrated scores.

Code references: `prepare_data.py`, `make_mfcc_frames.py`, `cv_subspace_pipeline.py`, `nsc_calibrated.py`.

3 Data & Features

Dataset composition: 5 classes; feature dimension D =60 (MFCC-20 + Δ + $\Delta\Delta$). Target sample rate 22.05 kHz, mono; frames from voiced audio with `frame_length=2048`, `hop_length=512`. The 60-D setup is used throughout.

Table 1: Dataset summary (per-fold averages).

Class	#train	#test	median frames/clip
V6	43.2	10.8	983.5
V8	48.0	12.0	992.5
inline-4	48.0	12.0	995.0
inline-6	48.0	12.0	998.0
single-cylinder	47.2	11.8	985.0

Artifacts: `../Results/cv/engine_configuration/fold_*/coverage.json`, `../Results/cv/engine_configuration/summary/table_A_lowdim.csv`, `../Results/summary/frames_per_clip_by_class.csv`.

Representation choice: MFCC vs. FFT. We adopt MFCCs rather than raw FFT magnitudes because MFCCs (i) apply a *mel-scaled filterbank* that smooths narrowband harmonics and noise into a perceptually weighted envelope, (ii) use *log compression* to stabilize dynamic range, and (iii) decorrelate via the DCT so that a small rank captures most variance. Empirically, PCA fitted on MFCC-60 exhibits markedly stronger low-rank compressibility than PCA on linear-frequency FFT magnitudes (513-D), whether using raw magnitude or log-magnitude.

Table 2: Low-rank compressibility across representations (cumulative EVR and rank needed for coverage).

Representation	D	**EVR@5**	**EVR@10**	k **for 90%**	k **for 95%**
MFCC+Δ+$\Delta\Delta$	60	91.9 %	96.5 %	5	8
FFT_mag	513	65.5 %	76.6 %	32	60
FFT_logmag	513	66.3 %	72.4 %	74	128

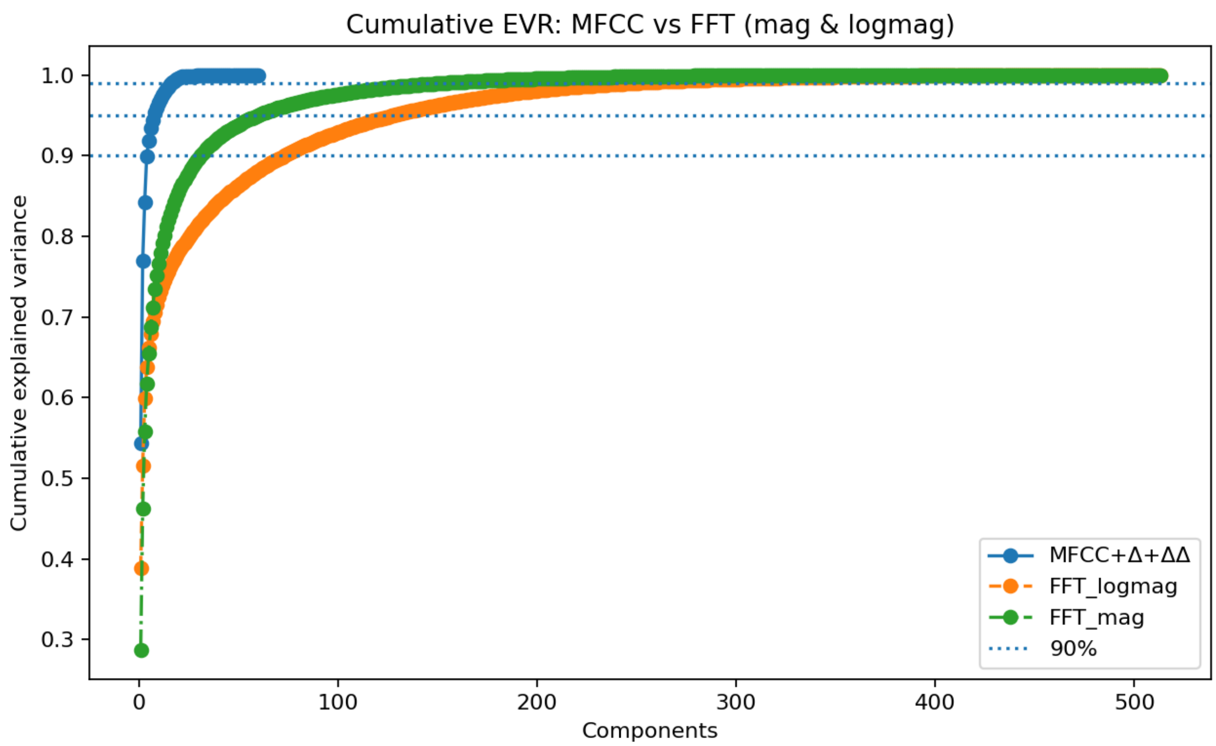
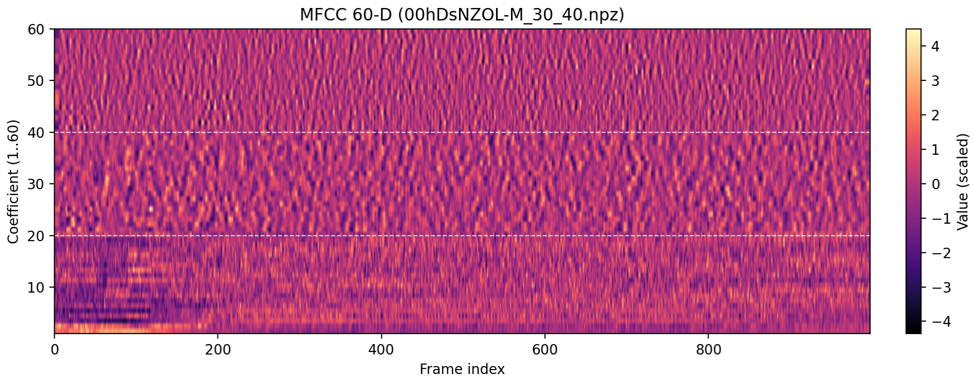


Figure 1: Cumulative explained variance (EVR) curves comparing PCA on MFCC-60 vs. FFT magnitudes (513-D) with and without log compression. MFCCs rise steeply, indicating stronger low-rank structure; FFT variants require substantially higher rank to reach the same coverage.

MFCC illustration and example. MFCCs summarize the short–time spectral envelope by projecting the log–mel spectrum onto cosine bases; first–order (Δ) and second–order ($\Delta\Delta$) differences emphasize dynamics. Below, we show a MFCC matrix (columns are frames; rows are coefficients).



(a) MFCC (+ Δ, $\Delta\Delta$)

Figure 2: Example features from a 10 s engine clip (22.05 kHz, `frame_length=2048`, `hop_length=512`). The MFCC panel stacks static coefficients with Δ and $\Delta\Delta$ (total D =60).

4 Methods (Subspace Modeling & Classification)

Preprocessing and MFCC features

Let $y[n]$ be a mono signal at $f_s = 22.05\,\text{kHz}$. We compute short-time spectra with an STFT using window $w[n]$, length N and hop H:

$$X[k,t] \;=\; \sum_{n=0}^{N-1} y[n+tH]\,w[n]\,e^{-j2\pi kn/N}, \qquad k = 0,\ldots,\lfloor N/2\rfloor. \tag{1}$$

Let $|X[k,t]|^2$ denote power. A mel filterbank $F\in\mathbb{R}^{M\times K}$ ($K{=}\lfloor N/2\rfloor{+}1$) maps linear frequencies to M mel bands:

$$S_{\text{mel}}[m,t] \;=\; \sum_{k=0}^{K-1} F[m,k]\,|X[k,t]|^2, \quad m = 0,\ldots,M{-}1. \tag{2}$$

We apply log compression (natural log) with small $\epsilon{>}0$ for numerical stability:

$$L[m,t] \;=\; \log\big(S_{\text{mel}}[m,t] + \epsilon\big). \tag{3}$$

MFCCs are the DCT-II of each log–mel vector (row-wise), keeping the first C coefficients:

$$c_q[t] \;=\; \sum_{m=0}^{M-1} D_{q,m}\,\big(L[m,t] - \bar{L}[m]\big), \quad q = 0,\ldots,C{-}1, \tag{4}$$

where D is the DCT-II matrix and $\bar{L}[m]$ is an optional per-band mean (cepstral mean subtraction). We then append first- and second-order regressors (deltas) using window W:

$$\Delta c_q[t] = \frac{\sum_{i=1}^{W} i\,(c_q[t+i] - c_q[t-i])}{2\sum_{i=1}^{W} i^2}, \tag{5}$$

$$\Delta^2 c_q[t] = \frac{\sum_{i=1}^{W} i\,(\Delta c_q[t+i] - \Delta c_q[t-i])}{2\sum_{i=1}^{W} i^2}. \tag{6}$$

We form frame vectors $x_t \in \mathbb{R}^D$ by stacking C MFCCs with Δ and $\Delta\Delta$ ($D = 3C = 60$). Per clip, we apply cepstral mean normalization (CMN): $x_t \leftarrow x_t - \frac{1}{T}\sum_{t=1}^{T} x_t$.

Class-conditional PCA subspaces

For each class c, collect TRAIN frames $\{x_t\}_{t\in\mathcal{T}_c}$ from its TRAIN clips and center by the class mean $\mu_c = \frac{1}{|\mathcal{T}_c|}\sum_{t\in\mathcal{T}_c} x_t$. Form $X_c = [\,x_t - \mu_c\,] \in \mathbb{R}^{D\times N_c}$ and compute its thin SVD

$$X_c \;=\; U_c\,\Sigma_c\,V_c^\top. \tag{7}$$

The rank-r subspace basis is $U_c^{(r)} = U_c[:, 1\!:\!r]$. The cumulative explained variance ratio (EVR) at rank k is

$$\mathrm{EVR}_c(k) \;=\; \frac{\sum_{i=1}^{k} \sigma_{c,i}^2}{\sum_{i=1}^{\min(D,N_c)} \sigma_{c,i}^2}. \tag{8}$$

Unless stated otherwise, we use a uniform $r = 5$ per class and log EVR and scree curves per fold.

Between-class geometry: principal angles

Given two subspaces with orthonormal bases $U \in \mathbb{R}^{D\times r}$ and $V \in \mathbb{R}^{D\times r}$, the cosines of the principal angles $\{\theta_i\}_{i=1}^{r}$ are the singular values of $U^\top V$:

$$U^\top V \;=\; P\,\mathrm{diag}(\cos\theta_1,\ldots,\cos\theta_r)\,Q^\top, \quad 0 \le \theta_1 \le \cdots \le \theta_r \le \tfrac{\pi}{2}. \tag{9}$$

We report the *largest* principal angle $\theta_{\max} = \theta_r$ as a summary separation (larger $\Rightarrow$ more separated).

Stability via bootstrap

For each class c, we assess subspace stability by bootstrapping TRAIN *clips*. In each of B replicates, sample a fraction p of the class's TRAIN clips without replacement, refit PCA to get $\tilde{U}_c^{(r)}$, and compute

$$\theta_{\max}^{(b)}(c) \;=\; \arccos\big(\sigma_{\min}(U_c^{(r)\top}\tilde{U}_c^{(r)})\big), \tag{10}$$

where $\sigma_{\min}(\cdot)$ is the smallest singular value. We summarize $\{\theta_{\max}^{(b)}(c)\}_{b=1}^{B}$ via median and IQR (and, when needed, bootstrap CIs).

Calibrated nearest-subspace classifier (NSC)

For a test clip with frames $\{x_t\}_{t=1}^{K}$ and class c with $(\mu_c, U_c^{(r)})$, define the framewise squared residual

$$r_{c,t} \;=\; \big\|\big(I - U_c^{(r)} U_c^{(r)\top}\big)\big(x_t - \mu_c\big)\big\|_2^2. \tag{11}$$

Aggregate over frames with an upper-tail trimmed mean (robust to outliers):

$$\tilde{r}_c \;=\; \operatorname{trimmean}\big(\{r_{c,t}\}_{t=1}^{K};\, q\big), \tag{12}$$

which discards the largest q fraction (we use $q = 0.40$ when $K \geq 10$, otherwise the median). Because residual scales can differ by class, we apply per-class z-score calibration estimated on TRAIN:

$$z_c \;=\; \frac{\tilde{r}_c - m_c}{s_c}, \qquad m_c = \mathbb{E}_{\text{TRAIN}}[\tilde{r}_c], \;\; s_c = \sqrt{\operatorname{Var}_{\text{TRAIN}}[\tilde{r}_c]}. \tag{13}$$

Predict the label by minimum calibrated score: $\hat{y} = \arg\min_c z_c$.

Inverting MFCCs to spectra (for interpretability)

Given a mean MFCC vector $\bar{c}$ (or a low-dimensional trajectory along principal directions), we reconstruct an approximate log–mel spectrum via the inverse DCT:

$$\widehat{L} \;=\; D^{-1}\bar{c}, \tag{14}$$

then undo log compression: $\widehat{S}_{\text{mel}} = \exp(\widehat{L}) - \epsilon$. To obtain a linear-frequency spectrum, use a nonnegative least-squares or pseudoinverse mapping of the filterbank:

$$\widehat{P} \;\approx\; F^{+}\,\widehat{S}_{\text{mel}}, \qquad F^{+} = \arg\min_{G} \|FG - I\|_F. \tag{15}$$

We visualize $\widehat{S}_{\text{mel}}$ or $\widehat{P}$ (linear frequency). If a time-domain signal is desired, a phase-reconstruction algorithm (e.g., Griffin–Lim) can be applied to $\sqrt{\widehat{P}}$, but in this work we use inversion only for spectral visualization.

Cross-validation and leakage control

We use 5-fold **StratifiedGroupKFold** with $y =$ `engine_configuration` and *groups* equal to the clip identifier, ensuring no frames from a test clip appear in train (no leakage). All preprocessing choices that can leak information (means, scales, calibration statistics) are fit on TRAIN within each fold and applied to TEST.

Uncertainty & significance. For accuracy metrics (overall, macro) we report the mean and standard deviation across the 5 cross-validation folds. For subspace stability we use a clip-level subsampling bootstrap on the TRAIN set within each class: on each replicate we sample a fraction $p = 0.70$ of training clips *without replacement*, refit the class PCA, and measure the largest bootstrap self-angle; we summarize these angles by the median and interquartile range (IQR) over $B_{\text{boot}} = 1000$ replicates. EVR and reconstruction metrics are likewise summarized by fold mean ± SD.

For significance of accuracy, we run a *one-sided* permutation test within each fold: shuffle the TEST labels by clip, recompute overall accuracy, and repeat $B_{\text{perm}} = 10{,}000$ times. The per-fold p-value is

$$p_f = \frac{1 + \#\{\hat{a}_f^{(b)} \geq a_f^{\text{obs}}\}}{1 + B_{\text{perm}}}, \tag{16}$$

and we combine $\{p_f\}_{f=1}^{5}$ across folds via Fisher's method,

$$X^2 = -2 \sum_{f=1}^{5} \ln p_f \;\sim\; \chi^2_{10}.$$

We additionally report a pooled permutation p for the mean accuracy across folds by comparing the observed mean to the distribution of fold-wise permuted means.

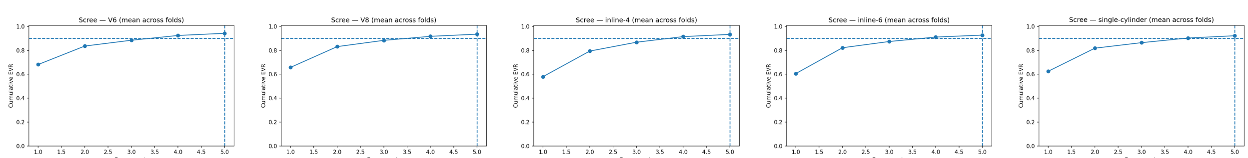
Defaults. $D = 60$, $r = 5$, $q = 0.40$, $K = 10$, $B_{\text{boot}} = 1000$, $p = 0.70$, $B_{\text{perm}} = 10{,}000$, 5 folds; seeds: CV=0, numeric=42.

5 Results

5.1 Low-Dimensionality

Table 3: EVR@5 (mean ± SD across folds).

Class	EVR@5 (mean ± SD)
V6	**94.3% ± 0.2%**
V8	**93.6% ± 0.5%**
inline-4	**93.4% ± 0.6%**
inline-6	**92.8% ± 0.5%**
single-cylinder	**92.3% ± 0.4%**



(a) V6 (b) V8 (c) inline-4 (d) inline-6 (e) single-cylinder

Figure 3: Representative scree plots (cumulative EVR) with r =5 marked. Fold chosen by median overall accuracy.

5.2 Interpretability via Subspace-to-Spectrum Inversion

To make the learned subspaces *audible* and visually interpretable, we invert class means and their principal directions from MFCC space back to spectra. Concretely, for each engine configuration we (i) take the class mean and top-r PCA directions in MFCC space, (ii) synthesize log-mel spectra via the MFCC inverse pipeline (pseudo-inverting the mel filterbank as needed), and (iii) optionally render linear-frequency views. We then overlay the per-class *mean* spectra to compare spectral envelopes across configurations.

What is overlaid in Fig. 4? Unless otherwise noted, the cross-class overlay uses the *PCA-filtered mean* curve, i.e., frames are first projected onto the learned class subspace ($P = UU^\top$ around the class mean μ with top-r PCs), then inverted to mel/STFT, and finally averaged (`-overlay_curve pca_filtered_mean`). Curves are normalized at a reference bin of 200 Hz (`-normalize refhz -ref_hz 200`) to enable shape comparisons. For reference, we also compute two non-subspace baselines that we do *not* call inversions: *model mean* (invert the tiled class-mean MFCC once) and *data mean* (average mel directly before mel→STFT).

Figure 4 shows the overlay of mean inversions. We also summarize simple descriptors computed on the inverted mean spectra: spectral centroid and band energies (50–200 Hz, 200–800 Hz, 0.8–3 kHz). Using these, the centroids rank as Inline-4 $\approx$ 2.81 kHz $<$ Inline-6 $\approx$ 3.32 kHz $<$ V6 $\approx$ 3.34 kHz $<$ V8 $\approx$ 3.45 kHz $<$ Single-Cylinder $\approx$ 3.72 kHz. In the 0.8–3 kHz band, Inline-4 and Inline-6 allocate the largest shares ($\approx$ 36.4% and 35.4%), followed by V6 (34.7%), V8 (32.0%), and Single-Cylinder (28.1%). Together these patterns indicate that inline configurations skew brighter in the mid–high band, while V-configurations are somewhat rounder; the single-cylinder mean carries comparatively more energy above 3 kHz (raising its centroid) but a smaller share in 0.8–3 kHz.

The leading principal direction acts as a *brightness axis*: moving along PC1 from "minus" to "plus" shifts the spectral centroid from roughly 0.6–1.2 kHz (PC1$-$) up to 3.7–6.0 kHz (PC1+), i.e., about -2.1 to -2.9 kHz below the mean and $+0.7$ to $+2.3$ kHz above the mean depending on class. Higher PCs tend to reallocate energy between lower and upper bands rather than uniformly brightening. Finally, the inverted model spectra track the empirical class-mean spectra closely in *shape* (Pearson $r \in [0.96, 0.99]$), confirming that inversion preserves the dominant envelope structure even if absolute amplitude differs due to normalization.

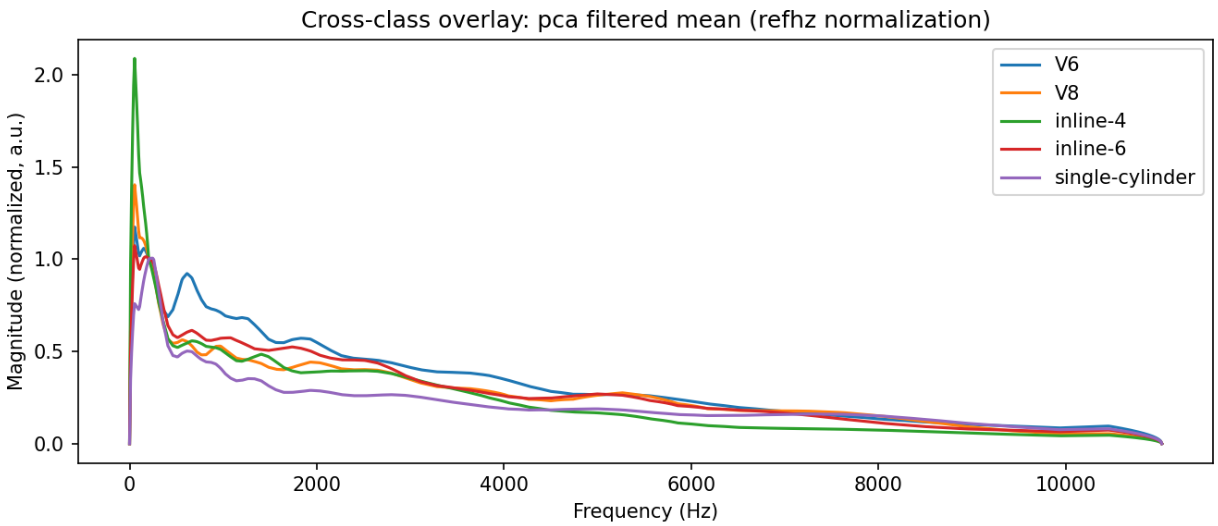


Figure 4: Subspace-to-spectrum inversion (engine configuration): cross-class overlay of *PCA-filtered mean* spectra (project → invert → average), with reference-Hz normalization at 200 Hz. Inline configurations concentrate slightly more energy in 0.8–3 kHz, whereas V-configurations are rounder; single-cylinder shows the highest centroid due to stronger upper-band content.

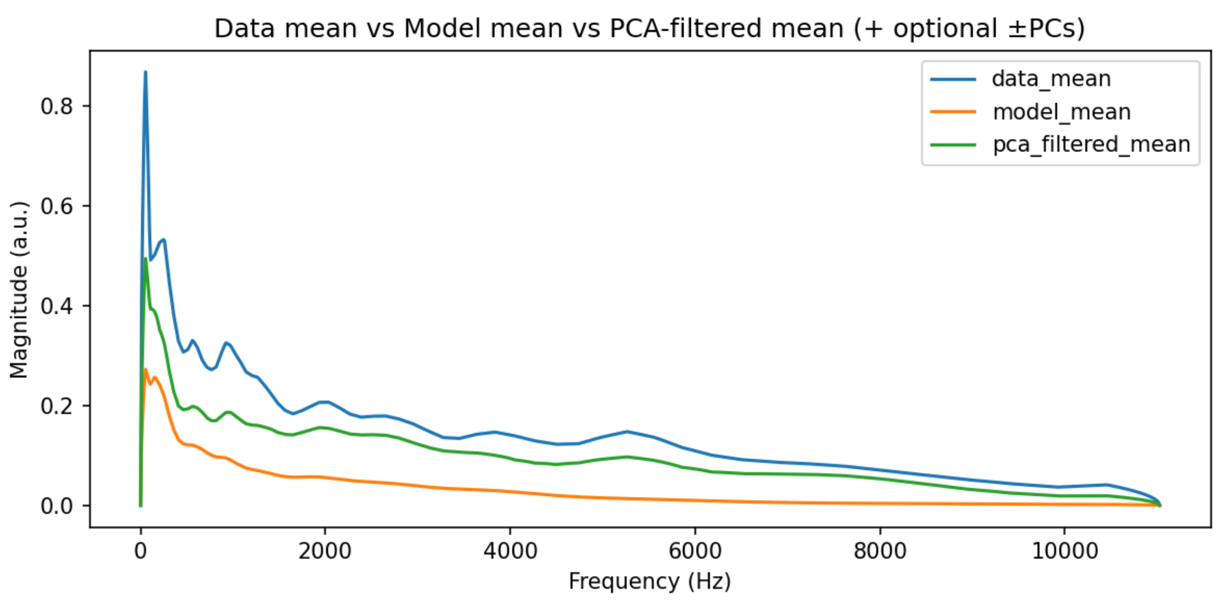
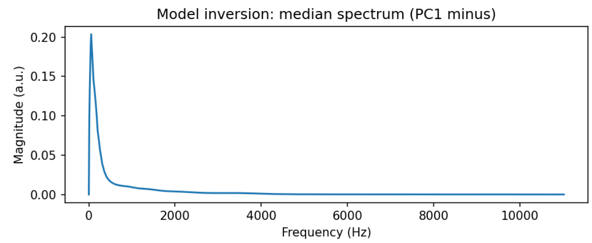
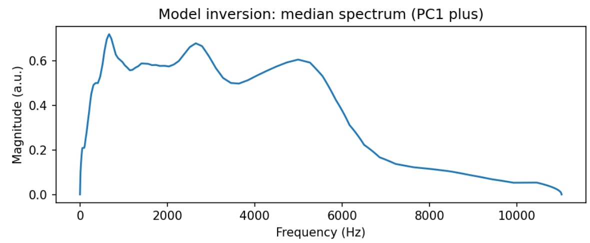


Figure 5: Per-class comparison (V8): *data mean* (no PCA), *model mean* (invert mean MFCC once), and *PCA-filtered mean* (project → invert → average). The PCA-filtered mean better matches the envelope while suppressing out-of-subspace variance, motivating its use in the cross-class overlay.



PC1− (Inline-4)



PC1+ (Inline-4)

Figure 6: Principal-direction excursions (Inline-4): PC1 behaves like a “brightness” control, shifting energy between low and high bands while preserving the overall envelope shape.

5.3 Stability

Table 4: Stability of class subspaces (largest principal angle, degrees).

Class	**Median**	**IQR (25–75%)**
V6	**14.1°**	10.8°–18.7°
V8	**13.5°**	10.3°–18.2°
inline-4	**13.1°**	9.6°–18.9°
inline-6	**26.0°**	16.8°–41.6°
single-cylinder	**13.1°**	10.4°–17.1°

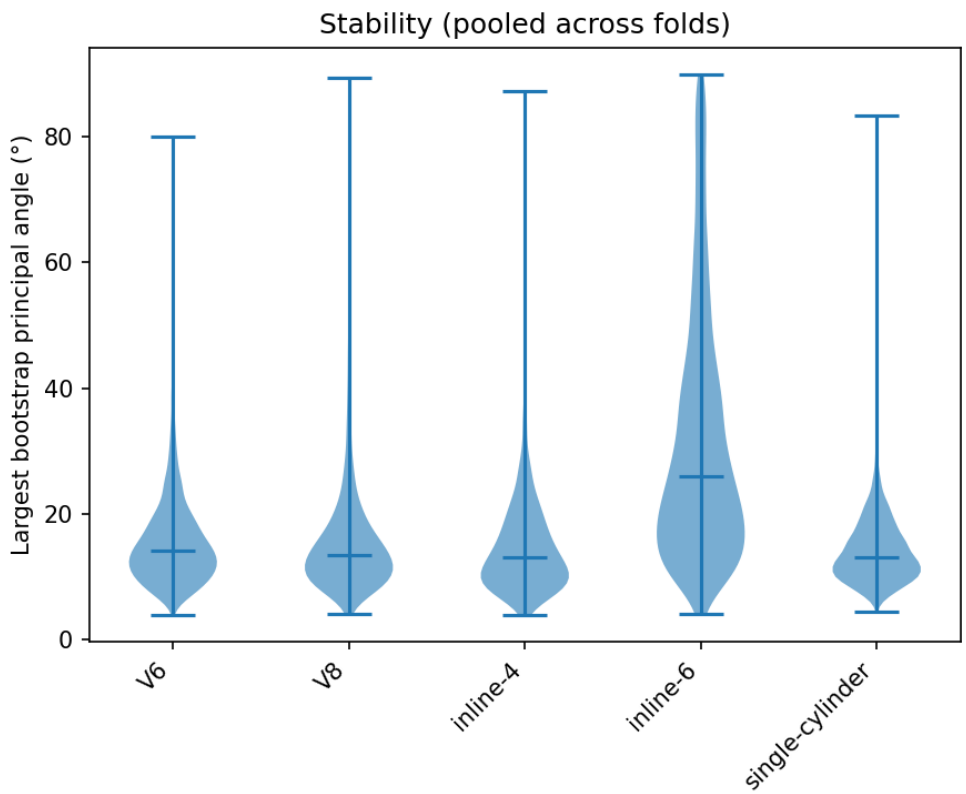


Figure 7: Stability distributions (violin of bootstrapped largest angles in degrees). Lower medians and tighter IQRs indicate more stable subspaces.

5.4 Between-Class Geometry

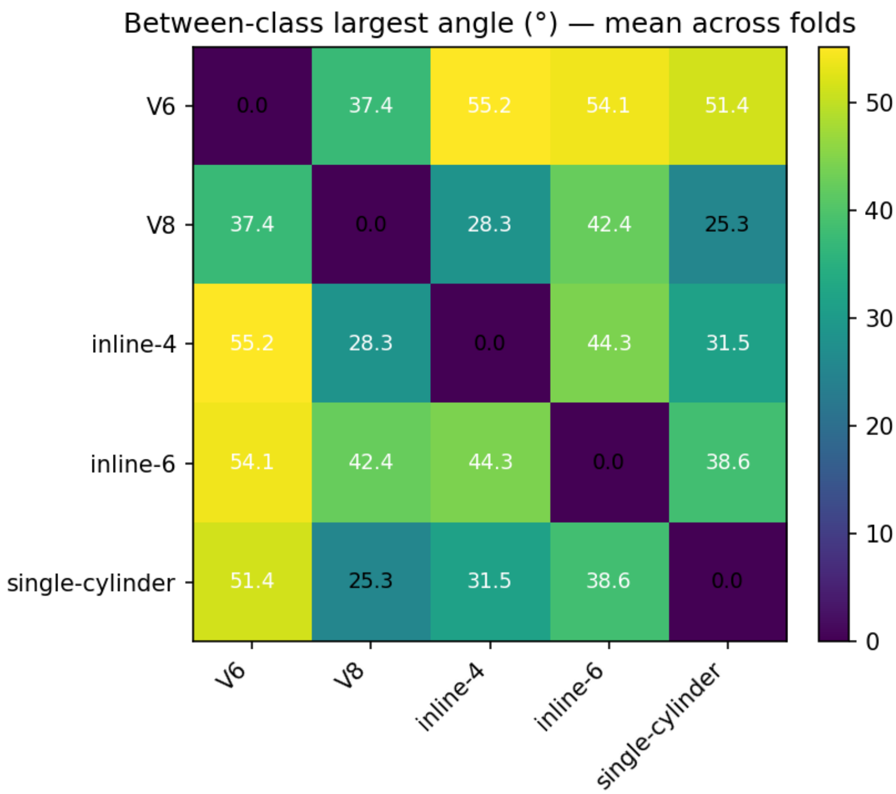


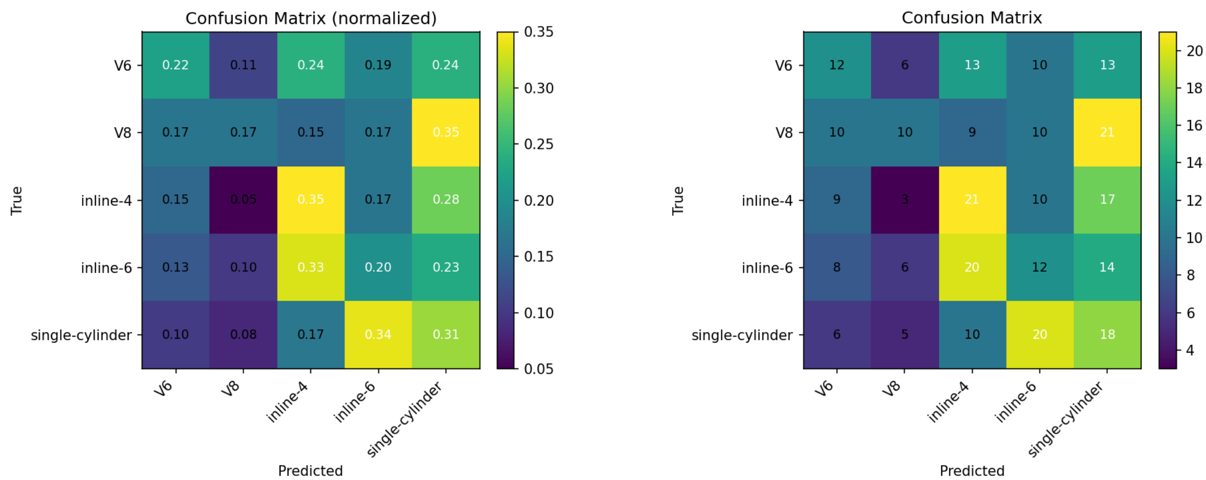
Figure 8: Between-class largest principal angles (degrees), mean across folds. Closest pair: V8 vs. single-cylinder (25.3°); most separated: V6 vs. inline-4 (55.2°).

5.5 Discriminativeness (NSC)

Table 5: NSC accuracy across folds. Chance baseline: $1/5 = 20\,\%$.

Fold	**Overall**	**Macro**
0	**0.136**	**0.141**
1	**0.271**	**0.284**
2	**0.237**	**0.239**
3	**0.328**	**0.329**
4	**0.276**	**0.252**
mean ± SD	**0.250 ± 0.071**	**0.249 ± 0.069**

Permutation tests (across folds). Combining one-sided per-fold permutation p-values using Fisher's method (Eq. 16) yields $\chi^2_{10} = 20.44$, $p = 0.0253$, indicating accuracy above chance with a small effect. A pooled permutation on the mean accuracy across folds gives $p = 0.0136$ (10,000 shuffles



(a) Normalized

(b) Raw

Figure 9: Confusion matrices aggregated across 5 folds (left: row-normalized; right: raw counts). Confusions align with closest pairs in angle space (e.g., V8 $\leftrightarrow$ single-cylinder).

per fold; observed mean 0.2495 vs. null mean 0.1990 ± 0.0227). JSON: `../Results/cv/engine_configuration/summary/perm_across_folds.json`.

6 Analysis & Interpretation

Low-dimensionality. Across all configurations, a uniform subspace rank of $r = 5$ (roughly 8% of the $D = 60$ feature dimension) captures between 92 % and 94 % cumulative explained variance (EVR), with negligible fold-to-fold variation ($\sigma_{\text{fold}} < 0.006$). Test-frame reconstruction errors remain moderate (≈ 1.1–1.3×10^3), confirming that five principal components are sufficient to represent the bulk of class variability. This demonstrates that engine-configuration audio indeed lies on low-dimensional manifolds within the MFCC space, with a mild weakness for inline-6 which exhibits slightly higher residuals.

Interpretability via inversion. Mean-spectrum inversions reveal physically sensible differences between configurations. The Inline-4 mean has the *lowest* centroid (≈ 2.81 kHz) and the *highest* 0.8–3 kHz share ($\approx$ 36.4%), consistent with a brighter mid–high emphasis, closely followed by Inline-6 (≈ 3.32 kHz, 35.4%). V6 and V8 sit in the middle (centroids ≈ 3.34 and 3.45 kHz; 0.8–3 kHz shares $\approx$ 34.7% and 32.0%), yielding rounder envelopes. Single-Cylinder exhibits the *highest* centroid (≈ 3.72 kHz) but the *lowest* 0.8–3 kHz share ($\approx$ 28.1%), indicating comparatively stronger content above 3 kHz. Principal-direction inversions clarify the geometry: PC1 reliably controls spectral "brightness," shifting centroids by roughly -2.1 to -2.9 kHz (PC1$-$) or $+0.7$ to $+2.3$ kHz (PC1+) relative to the mean. These trends align with expectations from source/cadence differences across engine families and demonstrate that the learned subspaces encode perceptually meaningful, class-specific envelopes rather than arbitrary variance.

Stability. Bootstrap principal-angle analysis indicates consistent subspace recovery for four classes: V6 (14.1°, 10.8°–18.7°), V8 (13.5°, 10.3°–18.2°), inline-4 (13.1°, 9.6°–18.9°), and single-cylinder (13.1°, 10.4°–17.1°). Inline-6 remains an outlier with a higher median (26.0°) and a much wider IQR (16.8°–41.6°), suggesting greater within-class heterogeneity (e.g., broader RPM range or recording variability). Aside from inline-6, the relatively narrow IQRs support that the PCA subspaces are reasonably stable under bootstrap resampling.

Between-class geometry. Pairwise subspace angles reveal a structured topology. The closest pair is V8–single-cylinder (about 25°), with V8–inline-4 moderately close (about 28°). Inline-6 sits farther from both (42°–44° to V8/inline-4), while V6 is the most distinct overall (37°–55° to others). This geometry aligns with the inversion patterns: classes with similar spectral envelopes produce smaller angles, whereas the more distinctive V6 envelope yields larger separations, indicating genuine acoustic relationships rather than random dispersion.

Discriminativeness. The calibrated nearest-subspace classifier (NSC) achieves overall accuracy of 25.0(71) % (macro 24.9(69) %), exceeding the 20 % chance baseline. Combining one-sided per-fold permutation p-values via Fisher's method (Eq. 16) yields $\chi^2_{10} = 20.44$, $p = 0.0253$; a pooled permutation test on the mean accuracy across folds gives $p = 0.0136$. V8 and inline-4 exhibit the strongest recall, while V6 and inline-6 lag, consistent with their instability and geometric proximity to other classes. These outcomes align with the subspace-angle structure: closer, less stable subspaces incur more confusions, whereas better-separated, stable subspaces yield higher recall.

Overall interpretation. Collectively, the results show that engine-configuration sounds inhabit low-dimensional, physically interpretable, and largely stable subspaces that preserve the perceptual hierarchy of engine types. However, overlapping manifolds and within-class heterogeneity limit discriminative performance, suggesting that finer state-conditioning (e.g., by RPM or engine load) or class-adaptive ranks could further separate the configurations. The analysis thus supports the hypothesis that audio from distinct engine architectures occupies compact but partially overlapping linear regions in MFCC space.

Reproducibility. All experiments use MFCC-20+Δ+$\Delta\Delta$ ($D = 60$) at 22.05 kHz, frames $(2048, 512)$. Per-class PCA with uniform $r = 5$ (chosen since EVR $\gtrsim 90\,\%$ by $k = 5$ for all classes). Stability via bootstrap principal angles ($B = 1000$, $p = 0.70$ of TRAIN clips), geometry via inter-class principal angles, and clip-level NSC with upper-tail trimming ($q = 0.40$, $K \geq 10$) and per-class z-calibration on TRAIN. CV: StratifiedGroupKFold(n=5), seeds `cv` = 0, `num` = 42. Artifacts (tables/figures/audio) are under `../Results/...`.

Synthesis. Taken together, the evidence indicates that engine-configuration audio occupies compact, interpretable subspaces (PC1 $\approx$ "brightness," PC2/PC3 $\approx$ spectral tilt/band emphasis). Inter-class geometry explains the limits of NSC: class pairs with small principal angles account for most confusions, whereas better-separated, more stable subspaces yield higher recall. Stability is strong for most classes; inline-6 is less stable, consistent with greater within-class heterogeneity. Overall discriminativeness is modest—25.0 % (7.1) vs. 20 % chance—statistically above chance but a small practical lift. Practically, conditioning on operating state (e.g., RPM) and allowing modest class-adaptive ranks should widen angles and improve discrimination without sacrificing interpretability; the subspace view remains valuable for compact indexing and diagnostic insight, especially when paired with stronger discriminative models.

7 Limitations & Future Work

Heterogeneity: Recording conditions and engine states may blur subspace boundaries. **Overlap:** Some class pairs have small between-class angles $\rightarrow$ systematic confusions.

Next steps: (i) Add baselines (SVM/kNN/CCR) and r-sweep; report CIs. (ii) Explore state-conditioned subspaces and mixture-of-subspaces, and Mahalanobis-weighted residuals. (iii) Integrate simple SNR weighting in frame aggregation.

Appendix

A. File Inventory (Python modules)

`prepare_data.py`: build balanced subsets; resample/trim; write `Data/` metadata and frames.
`make_mfcc_frames.py`: compute MFCC-20+Δ+$\Delta\Delta$ ($D = 60$); write `Data/mfcc` and index parquet.
`cv_subspace_pipeline.py`: 5-fold CV for low-dimensionality, stability, geometry, and NSC; writes `Results/cv/...`.
`nsc_calibrated.py`: calibrated nearest-subspace with trimmed aggregation ($q = 0.40$, $K \geq 10$).
`nsc_eval.py`: evaluation utilities (non-CV).
`split_pca_per_class.py`: per-class PCA and scree saves.
`pairwise_subspace_angles.py`: principal-angle geometry utilities.
`subspace_stability_bootstrap.py`: bootstrap stability analysis.

B. Selected per-fold summaries

See ../Results/cv/engine_configuration/fold_*/reconstruction_mse.csv, ../Results/cv/engine_configuration/fold_*/stability_summary.csv, ../Results/cv/engine_configuration/fold_*/between_class_angles.csv, ../Results/cv/engine_configuration/fold_*/nsc_accuracy.json.

C. Full confusion matrices per fold

See ../Results/cv/engine_configuration/fold_*/confusion_raw.png and ../Results/cv/engine_configuration/fold_*/confusion_norm.png.

D. Raw stability angle samples

See ../Results/cv/engine_configuration/fold_*/stability_raw.csv for per-bootstrap samples.